Non-Degeneracy and Reality of Bound States in 1D

Source: D. Tong, Quantum Mechanics, Problem Sheet 1, Question 3.

Consider a particle in one dimension in a potential $U(x)$ that tends to zero rapidly as $x \to \pm\infty$.

1. Let $\chi_1(x)$ and $\chi_2(x)$ be normalisable energy eigenfunctions of the Hamiltonian with the same energy eigenvalue E. By considering $\chi_1\chi_2' - \chi_2\chi_1'$, show that χ_1 and χ_2 must be proportional to one another. What does this mean, physically?
2. Can the wavefunction for a normalised energy eigenstate always be chosen to be real?
3. Show that if $\chi(x)$ is any normalised energy eigenstate then $\langle \hat{p} \rangle_\chi = 0$.

Solution. Consider the Wronskian,

$$W(x) = \chi_1\chi_2' - \chi_2\chi_1'. \tag{1.1}$$

Note that

$$W'(x) = \chi_1\chi_2'' - \chi_2\chi_1''. \tag{1.2}$$

Since both χ_1 and χ_2 satisfy the Schrödinger equation with the same eigenvalue, $W'(x)$ vanishes. Therefore, we have $W(x) = \text{const}$. Since the wavefunctions are normalisable, we can only have $W(x) = 0$. Thus, the two wavefunctions are linearly dependent, which means they are the same eigenfunction to the Hamiltonian.

Given that χ satisfies

$$H\chi = E\chi, \tag{1.3}$$

take the complex conjugate on both side, we find

$$H\chi^* = E\chi^*. \tag{1.4}$$

As we have shown that χ and χ^* are linearly dependent, we can always construct a real eigenfunction for the eigenvalue, E, namely

$$\psi = \chi + \chi^*. \tag{1.5}$$

The expectation value for the momentum operator is given by (χ taken to be real)

$$\langle p \rangle = -i\hbar \int dx \chi\chi' = -\frac{i\hbar}{2} \int dx (\chi^2)' = -\frac{i\hbar}{2} [\chi^2(+\infty) - \chi^2(-\infty)], \tag{1.6}$$

which must vanish for normalisable χ.